

VIRTUAL DATA SCIENCE EXPLORER INTERNSHIP

Assignment Report

DATA SCIENCE PROJECT PLANNING **AND STRATEGY DESIGN**

PROJECT TITLE

Titanic Passenger Survival Prediction
using Data Science and Machine Learning

Submitted By

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Internship

Virtual Data Science Explorer

Internship Provider

YuvaIntern (via Skill India Digital Platform)

Project Phase

Week 1 – Project Planning

Tools

Python 3.11, Pandas, NumPy, Matplotlib, Seaborn,
Scikit-learn, VS Code, Git & GitHub

Academic Year 2026

Virtual Data Science Explorer Internship

Data Science Project Planning and Strategy Design

Project Title: *Titanic Passenger Survival Prediction using Data Science and Machine Learning*

1. Introduction

Data Science combines statistics, programming, and machine learning to discover patterns from data and support better decisions. This project proposes an end-to-end data science workflow using the Titanic dataset to demonstrate planning, analysis, visualization, machine learning, and reporting.

2. Project Background

The Titanic dataset contains passenger information such as age, gender, ticket class, fare, family relationships, and survival status. It is widely used to learn data science because it contains missing values, categorical variables, and meaningful relationships suitable for analysis.

3. Problem Statement

The objective is to identify the factors that influenced passenger survival and design a complete workflow that could later be implemented using Python. The final outcome will be a predictive model supported by data cleaning, visualization, and analysis.

4. Project Objectives

- Collect a publicly available dataset.
- Clean missing and inconsistent data.
- Perform Exploratory Data Analysis (EDA).
- Visualize important trends and relationships.
- Build and evaluate a machine learning model.
- Present findings in a professional report.

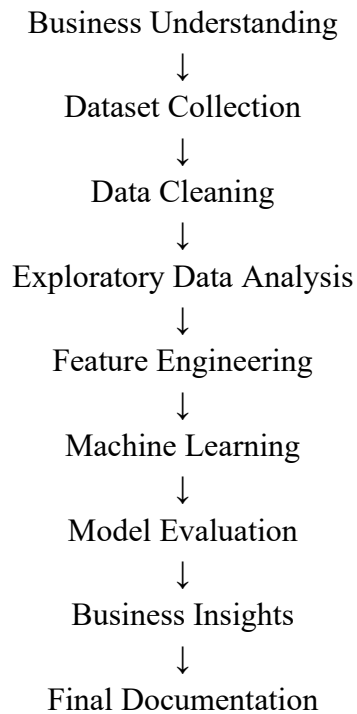
5. Project Scope

- In Scope
 - Data acquisition
 - Data cleaning
 - EDA
 - Statistical analysis
 - Machine learning
 - Final report

- Out of Scope
 - Web deployment
 - Mobile application
 - Real-time prediction

6. Methodology & Strategy

The project follows this workflow:



7. Tools and Technologies

- Python 3.11
- Visual Studio Code
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Microsoft Word
- Git and GitHub

8. Estimated Timeline

Task	Hours
Project Planning	4
Data Collection	2
Data Cleaning	6
EDA	8
Statistical Analysis	5
Machine Learning	8
Documentation	5

9. Expected Outcomes

- Clean and organized dataset.
- Meaningful visualizations.
- Identification of survival factors.
- Machine learning prediction model.
- Professional GitHub-ready project.
- Improved data science skills.

10. Challenges and Risk Mitigation

Challenge	Mitigation
Missing values	Use median/mode imputation
Duplicate records	Remove duplicates
Outliers	Detect using boxplots
Overfitting	Train/test split and validation
Feature selection	Correlation analysis

11. Expected Learning

This project will strengthen knowledge of Python programming, data preprocessing, exploratory data analysis, visualization, machine learning fundamentals, documentation, Git, and GitHub portfolio development.

12. Conclusion

This report presents the strategic plan for a complete data science project using the Titanic dataset. The planned workflow mirrors a real-world project lifecycle, beginning with project planning and ending with predictive modeling and professional reporting. The following internship tasks will implement each planned stage using Python.